Hrishikesh Belagali

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Education

Michigan State University — **Expected Graduation: Spring 2027**

Bachelor of Science in Computer Science, Minor in Mathematics — *4.0 GPA*

Relevant Coursework

Graduate Quantum Algorithms, Graduate Computational Foundations of Artificial Intelligence, Advanced Linear Algebra, Data Structures and Algorithms

Experience

Honors College — **August 2025 – Present**

Honors Research Scholar — *East Lansing, Michigan*

- Developed generalized algorithms for simulating quantum circuits using group homomorphisms and the stabilizer formalism.
- Validated theoretical results using numerical experiments in Python, NumPy and Qiskit.
- Analyzed circuit families to identify structural properties enabling reduced T-gate counts.
- Prepared research figures, proofs and write-ups for an upcoming first-author manuscript submission.

MSU-DOE Plant Research Laboratory — **December 2024 – Present**

Undegraduate Research Assistant — *East Lansing, Michigan*

- Investigated interactions between chloroplast targeting protein (At4g16060) and thylakoid membranes using molecular dynamics in NAMD.
- Analyzed the effect of PDLP-5 on water permeability through PIP1;4 aquaporin tetramers using grid-steered molecular dynamics simulations and TCL scripting.
- Conducted research on the effects of cyclopropane fatty acids on bacterial membrane structure and dynamics in the presence of commercially relevant biofuels.
- Set up analysis pipelines using Python, NumPy, MDAnalysis and Matplotlib to extract and visualize system properties.
- Prepared figures and write-ups for an upcoming first-author manuscript submission.

Projects

aqua-blue | *Python, NumPy*

Secondary maintainer of aqua-blue, a package building echo state networks.

- Implemented core reservoir computer hyperparameters including state time-lag, sparsity, and spectral radius.
- Added timezone-aware datetime support to streamline integration with data pipelines.

Discretized Quantum Adiabatic Evolution | *Python, Qiskit, NumPy*

- Implemented a quantum adiabatic evolution algorithm using the Trotter–Suzuki approximation for time discretization.
- Solved QUBO problems via quantum Ising models, improving solution convergence and stability.

Quantum Arithmetic Circuits | *Python, Qiskit, NumPy*

- Designed and implemented quantum circuits for addition, subtraction, and multiplication without classical overhead.
- Applied the Quantum Fourier Transform to reduce qubit requirements for arithmetic operations.

Protein Structure Prediction | *Python, MatPlotLib, NumPy, OVITO*

- Proposed a hybrid Monte Carlo algorithm to optimize Irbäck's off-lattice protein folding model.
- Accurately modelled protein folding within 1-4 RMSD.

Quadratic Unconstrained Binary Optimization Problems | *Python, MatPlotLib, NumPy*

- Developed a simulated annealing algorithm to determine the global minimum of QUBO problems.
- Improved convergence through the implementation of dynamic stochastic tunneling.

Technical Skills

Languages: Python, C++, C, LaTeX
Packages: NumPy, Qiskit, Cirq, PyTorch, SciPy, Matplotlib
Technologies/Frameworks: Git, GitHub, SLURM, CMake, Linux